MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

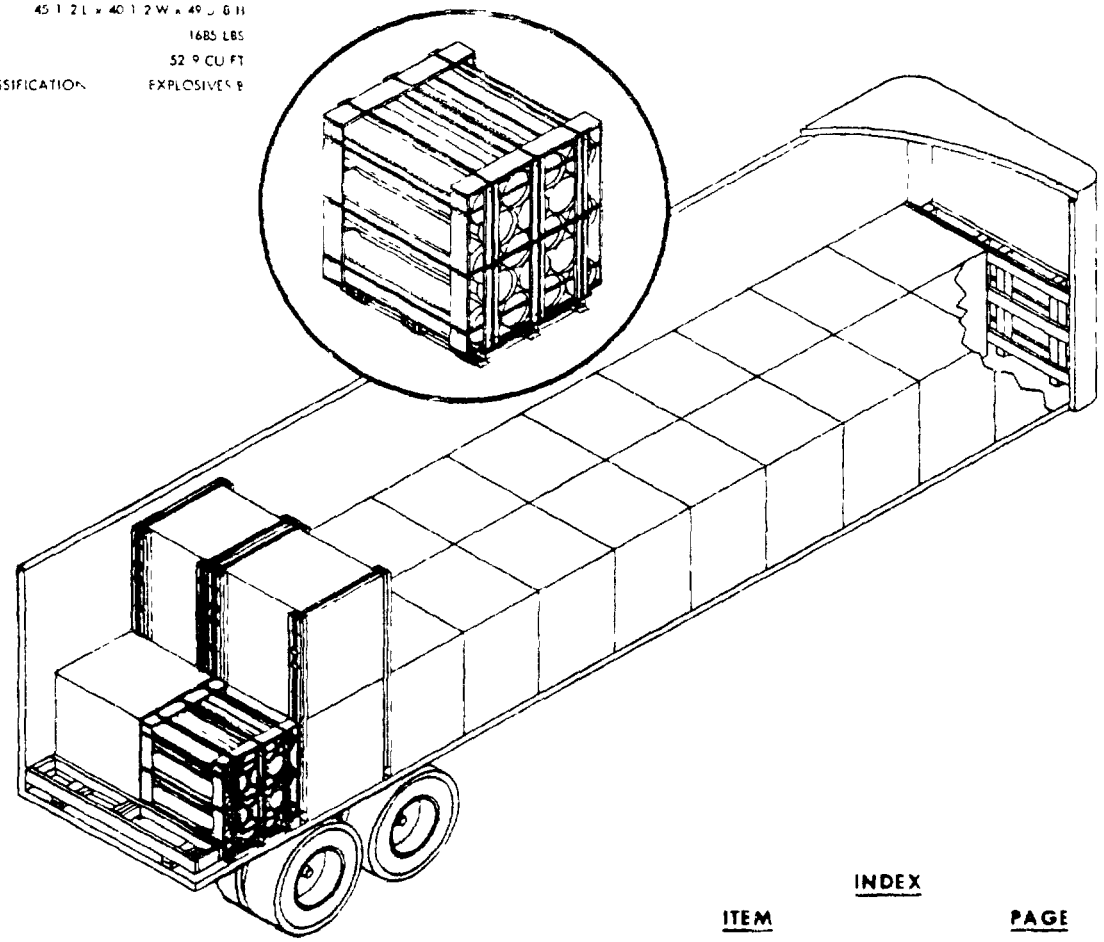
CHARGE,PROPELLING 6"/47
DOMESTIC UNIT LOAD

MIL-STD-1320-157
(NAVY)

10 SEPTEMBER 1974

UNIT LOAD DATA

UNIT LOAD DRAWING	WR 53/842
DIMENSIONS	45 1/2 L x 40 1/2 W x 49 3/8 H
WEIGHT (APPROX)	1685 LBS
CUBE	52.9 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES B



NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINER LOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40 42 & 44 FT TRAILER (VAN)	2 & 3
LTL 40 FT TRAILER (VAN	4 & 5
UNITIZING PROCEDURE	5

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M S Gray NWC 9/5/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J E Kelly 9/6/74
SIGNATURE SEASYSCOM BY DIRECTION DATE

ORIGINATOR L R Ramsey 9/5/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE PROPELLING CHARGE 6"47 IN CARTRIDGE TANK MK 4 WHEN PALLETIZED IN ACCORDANCE WITH DOMESTIC UNIT LOAD WP 53/B42

NOTE

TRAILER INSIDE WIDTH MUST BE 91 1/2 INCHES MINIMUM

2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42 & 44 FT VAN TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT EXTREME REAR OF THE TRAILER
3. THE LENGTH OF STRUT PIECE 3 SHALL BE SUFFICIENT TO CAUSE THE REAR BLOCKING ASSEMBLY DETAIL B TO BEAR AGAINST THE UNIT LOADS AND THE TRAILER DOORS WHEN THE TRAILER DOORS ARE CLOSED

WARNING

DO NOT USE TRAILERS WITH ROLL UP REAR DOORS WHEN BLOCKING AND BRACING BY THIS METHOD

4. TO PREVENT LONGITUDINAL MOVEMENT OF UNIT LOADS WHEN THERE IS A SECOND LAYER THE UNIT LOAD SHALL BE UNITIZED (SEE UNITIZING DETAIL PAGE 5 WHERE THE LOAD CHANGES FROM TWO LAYERS HIGH TO ONE LAYER HIGH STIFFENERS SHALL BE USED ON BOTH END OF THE STACKED TWO HIGH LOAD

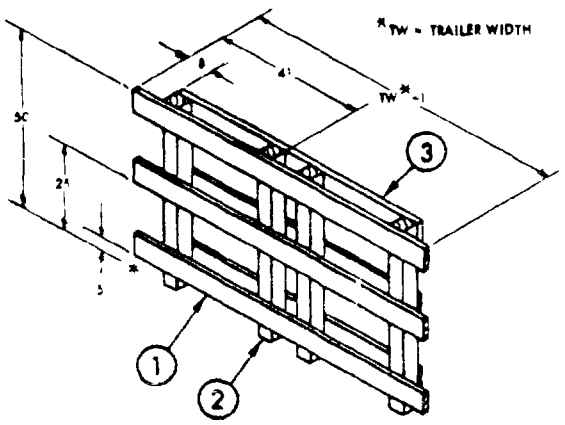
NOTE

WHEN UNIT LOADS ARE STACKED TWO HIGH THE INSIDE HEIGHT OF THE TRAILER MUST BE OVER 102 INCHES

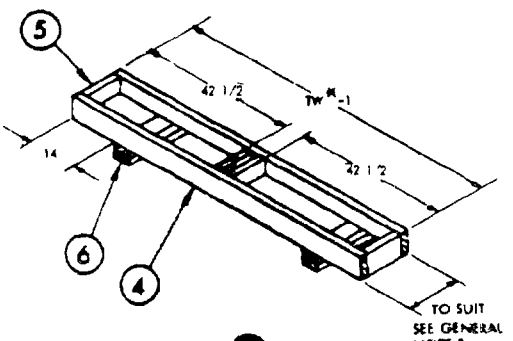
5. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS
6. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA CLOSE AND SEAL TRAILER DOORS AND ATTACH EXPLOSIVES B PLACARD TO BOTH SIDES FRONT AND BACK OF TRAILER
7. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER MM-L-751 NAILS FF-N-105 TYPE II STYLE 10 COMMON BRIGHT STRAPPING QQ-S-781 TYPE I, CLASS A PLYWOOD COMMERCIAL STANDARD CS 45 GRADE INTERIOR C-D
8. FOR GENERAL TRUCK LOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

FTL 40, [redacted] AND 44 FT TRAILER (VAN)

1. WHEN 44 FT OR 45 FT VANS ARE USED FOR FTL OF 24 UNIT LOADS IT IS NOT NECESSARY TO DOUBLE TIER OR UNITIZE USE DOUBLE ROW PATTERN AND BLOCK AND BRACE IN SAME MANNER AS FOR THE 40 FT TRAILER
2. WHEN A 40 FT VAN IS USED LOAD BLOCK AND BRACE AS SHOWN ON PAGE 3

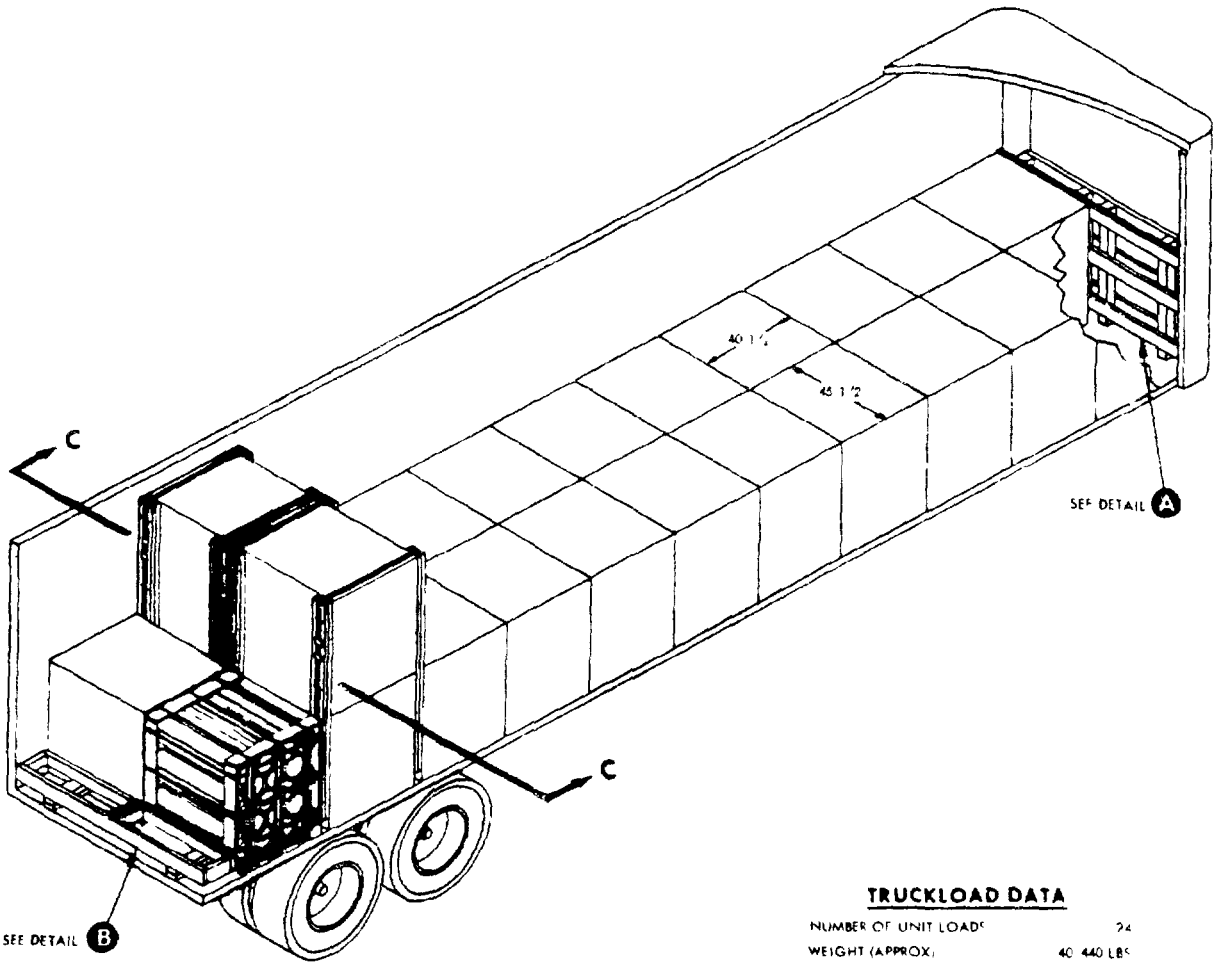


DETAIL A
FRONT BULKHEAD
1 REQUIRED



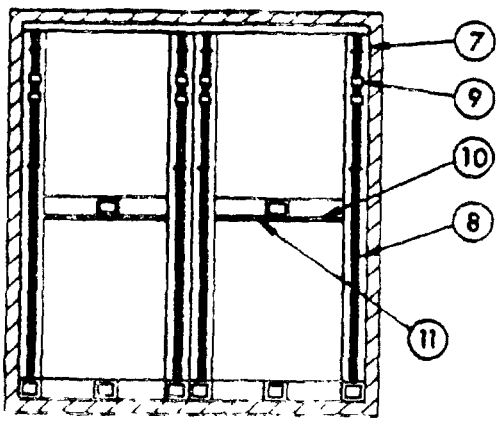
DETAIL B
REAR BLOCKING ASSEMBLY
1 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS	24
WEIGHT (APPROX)	40 440 LBS



SECTION C-C

*TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER NAILS	SIZE
11	PLYWOOD	40 x 45 1/2 x	-	SEE UNITIZING DETAIL PAG		
10	STAPLE	FOR 1 1/4 STRAP	32			
9	SEAL	FOR 1 1/4 STRAP	8			
8	STRAP	1 1/4 x .035 x 25 FT	4		-	
7	STIFFENER	2 x 6 x 94	16	DOUBLED & LAMINATED	4	10d
6	SUPPORT	2 x 4 TO SUIT	4	4	3 PER JOINT	10d
5	STRUT	4 x 6 TO SUIT	4	SEE 4	4	10d
4	CROSSMEMBER	2 x 6 x TW*			4 PER JOINT	16d
3	FWD CROSSMEMBER	2 x 6 TO SUIT		7	3 PER JOINT	12d
2	VERTICAL	4 x 4 x 50	4	SEE 1 &		
1	AFT CROSSMEMBER	2 x 6 x TW* 1	3	7	3 PER JOINT	12d

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.

LTL 40 FT TRAILER (VAN)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE

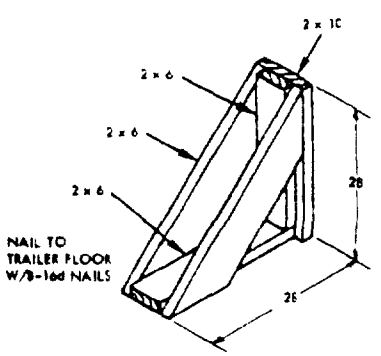
TABLE 1

NO. OF UNIT LOADS	LOAD PATTERN*	FWD
24		
22		
21		
20		

*LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER NUMBER IN BLOCKS INDICATES NUMBER OF UNIT LOADS IN LAYER

WHEN REQUIRED TO SHIP 23 UNIT LOADS USE LOAD PATTERN SHOWN IN TABLE 1 BLOCK AND BRACE LOAD AS SHOWN FOR FTL ON PAGE 3 THE SINGLE TWO HIGH STACK OF UNIT LOADS SHALL HAVE TWO ADDITIONAL STRAPS TO PREVENT LATERAL MOVEMENT AS SHOWN IN THE UNITIZING DETAIL PAGE 5

3. WHEN REQUIRED TO SHIP 22 UNIT LOADS USE LOAD PATTERN SHOWN IN TABLE 1 BLOCK AND BRACE LOAD AS SHOWN FOR FTL ON PAGE 3
4. WHEN REQUIRED TO SHIP 21 UNIT LOADS USE LOAD PATTERN SHOWN IN TABLE 1 BLOCK AND BRACE LOAD AS SHOWN FOR FTL ON PAGE 3 AND ADD 2 KNEE BRACES (DETAIL D) EACH AGAINST WOODEN FRAME OF UNIT LOAD
5. WHEN REQUIRED TO SHIP 20 UNIT LOADS USE LOAD PATTERN SHOWN IN TABLE 1 USE FRONT BULKHEAD AS SHOWN IN FTL PAGE 3 AND TWO KNEE BRACE DETAIL D PER ROW (4 PER TRAILER) EACH AGAINST WOODEN FRAME OF UNIT LOAD
6. WHEN REQUIRED TO SHIP 19 OR LESS UNIT LOAD USE DOUBLE ROW PATTERN BLOCK AND BRACE AS FOR 20 LOADS
7. ACTUAL SHIPPING REQUIREMENTS DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED THE LOAD PATTERN AND DUNNAGING SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS DOCUMENT AND THE GENERAL TRUCKLOADING STANDARD MIL-STD-1320 (NAVY)



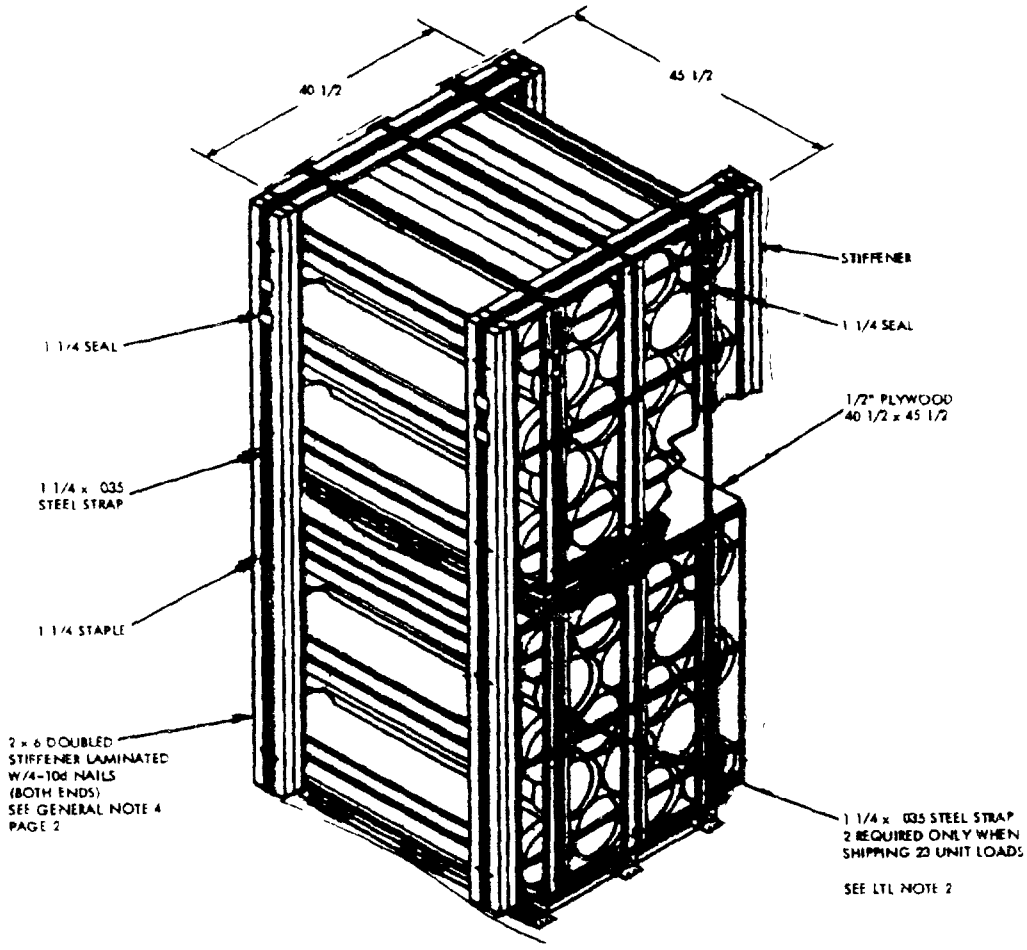
DETAIL (D)

KNEE BRACE
2 REQUIRED PER ROW
(4 REQUIRED PER DOUBLE ROW)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.

UNITIZING PROCEDURE

1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD THE TWO HIGH STACKED UNIT LOADS SHALL BE STRAPPED AS SHOWN
2. THE 2 x 6 DOUBLED STIFFENER SHALL EXTEND FROM THE TOP OF STACKED UNIT LOADS TO THE TOP OF THE PALLET OF THE BOTTOM UNIT LOAD. THE STIFFENER SHALL BE LAMINATED WITH FOUR 10d NAILS TO PREVENT SHIFTING
3. THE 1 1/4 INCH STEEL STRAPS POSITIONED AS SHOWN ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET WITH THE STRAPS HOLDING THE STIFFENERS IN PLACE
4. TENSION STRAPS AND SEAL WITH TWO DOUBLED CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLES



REVIEW ACTIVITIES
NAVY - OS, AS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. 81[illegible])

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:45Z
Check the source to verify that this is the current version before use.